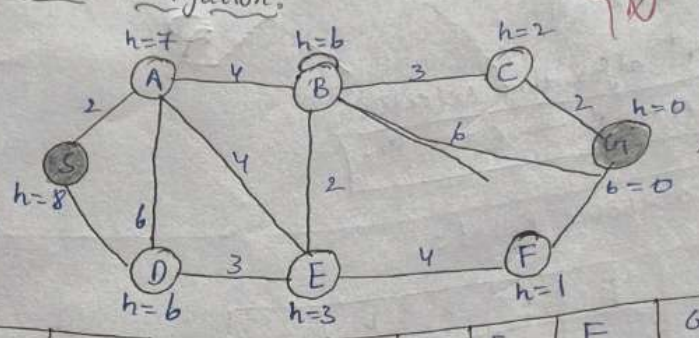


① Smart Ambulance Navigation:



Junction	S	A	B	C	D	E	F	G
$h(n)$	8	7	6	2	6	3	1	0

A* calculation:

$g(n)$ = actual cost from start to node n
 $h(n)$ = estimated cost from n to goal
 $f(n)$ = total estimated cost.

start at S:

$g(s) = 0, h(s) = 8$
 $f(s) = 0 + 8 = 8$

From S:

A:
 $g = 2, h = 7, f = 9$

D:
 $g = 6, h = 6, f = 12$

From A:

B:
 $g = 2 + 4 = 6, h = 6, f = 12$

E:
 $g = 2 + 4 = 6, h = 3, f = 9$

D: existing cost
 $g = 6, f = 12$

From E:

B: new
 $g = 8, f = 14$
 worse than existing
 $g = 6$

D: new
 $g = 9, f = 15$
 worse than existing
 $g = 5$

F:
 $g = 10, h = 1, f = 11$

select A because it has the lowest f .

select E, because $f(E) = 9$.

Continue with the lowest
 $B \rightarrow C \rightarrow G$ the goal is reached.

Selected route:

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

Total cost

$$2 + 4 + 3 + 2 = 11$$

$\therefore$ The A^* algorithm selects:

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

2. Autonomous warehouse Robot:

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

Ans) The warehouse robot does not initially know all obstacles. Therefore, an online search Agent with an informed search method such as online A^* is suitable.

Given grid

start

$S = (1, 5)$

Goal:

$G = (7, 1)$

Known obstacles:

$(2, 4), (5, 4), (2, 2), (6, 2), (4, 3), (3, 1)$

category:

The robot uses:

$$f(n) = g(n) + h(n)$$

with Manhattan distance as the heuristic:

$$h(n) = |x - x_g| + |y - y_g|$$

The robot repeatedly:

- 1) observes nearby cells.
- 2) Updates its map.
- 3) selects the best available next cell.
- 4) Moves to the cell.
- 5) Marks it as blocked
- 6) Recalculates the path.
- 7) continues until it reaches the goal.

one safe route

A valid route through the given grid is:
 $(1,5) \rightarrow (2,5) \rightarrow (3,5) \rightarrow (4,5) \rightarrow (5,5) \rightarrow (6,5) \rightarrow (7,5) \rightarrow (7,4) \rightarrow (7,3) \rightarrow (7,2) \rightarrow (7,1)$
Goal is $(1,7)$

online search vs offline search

online search	offline search
Environment is Partially unknown	complete environment is known
Discovers obstacles during movement	obstacles are known before movement
Replans when new information obtained	usually plans once
suitable for changing	suitable for completely known environments
uses sensing and learning	uses a pre-built map.

online A* is more suitable because the robot cannot depend on complete prior knowledge. it can see obstacles, update its map, & dynamically re-plan its route.

3. University Examination scheduling:

Ans) The problem requires six examinations to be assigned for four time slots while satisfying student-conflict and hall-capacity constraints.

a) CSP Formulation:

$$X = \{E_1, E_2, E_3, E_4, E_5, E_6\}$$

Each variable represents the time slot assigned to an examination.

Domains:

Initially:

$$D(E_1) = D(E_2) = D(E_3) = D(E_4) = D(E_5) = D(E_6) \\ = \{T_1, T_2, T_3, T_4\}$$

Student-conflict constraints:

From the question:

$$E_1 \neq E_2$$

$$E_1 \neq E_3$$

$$E_2 \neq E_4$$

$$E_3 \neq E_5$$

$$E_4 \neq E_6$$

$$E_2 \neq E_5$$

These are the six given conflict relationships,

Hall capacities:

Hall	Capacity
Hall A	120
Hall B	90
Hall C	60

Assignment 2

$E1 = T2$

E1 conflicts with E2 & E3:

$E3 \rightarrow \{T1, T3, T4\}$

Assignment 3

$E4 = T2$

$E6 \rightarrow \{T1, T3, T4\}$

Assignment 4

$E3 = T1$

$E5 \rightarrow \{T2, T3, T4\}$

Assignment 5

$E5 = T2$

Assignment 6

$E6 = T1$

$T1 \neq T2$

$\therefore$ This is valid.

c) Domain update Table

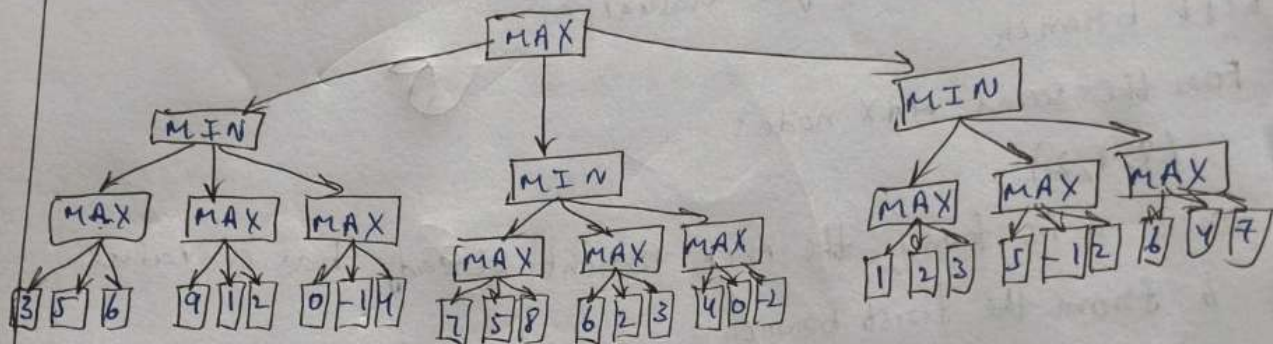
step	Assignment	Remaining important domains
Initial	—	$All = \{T1, T2, T3, T4\}$
1	$E2 = T1$	$E1, E4, E5 = \{T2, T3, T4\}$
2	$E1 = T2$	$E3 = \{T1, T3, T4\}$
3	$E4 = T2$	$E6 = \{T1, T3, T4\}$
4	$E3 = T1$	$E5 = \{T2, T3, T4\}$
5	$E5 = T2$	No conflict introduced
6	$E6 = T1$	All assignments valid

Final Timetable:

Time slot	Examinations	Hall Allocation
T1	E2, E3, E6	E2 → B, E3 → A, E6 → C
T2	E1, E4, E5	E1 → A, E4 → B, E5 → C
T3	-	-
T4	-	-

4.

Game - Playing AI:



Minimax calculation:

Left MIN branch

MAX nodes

1. $(3, 5, 6)$

MAX = 6

2. $(9, 1, 2)$

MAX = 9

3. $(0, -1, 4)$

MAX = 4

MIN chooses

$\text{MIN}(6, 9, 4) = 4.$

Right MIN branch

1. $(1, 2, 3)$

MAX = 3

2. $(5, -1, 2)$

MAX = 5

3. $(6, 4, 7)$

MAX = 7

Exam sizes are:

Exam	Students
E1	100
E2	80
E3	70
E4	60
E5	90
E6	50

These capacities and student numbers are given in the assignment.
Therefore:

- E1 $\rightarrow$ Hall A only
- E2 $\rightarrow$ Hall A or B
- E3 $\rightarrow$ Hall A or B
- E4 $\rightarrow$ Hall A, B (or) C
- E5 $\rightarrow$ Hall A or B
- E6 $\rightarrow$ Hall A, B (or) C

b) MRV + Forward Checking:

Choose E2 first using MRV/tie-breaking based on constraints

Assignment 1

E2 = T1

Forward checking removes T1 from its neighbors:

E1 $\rightarrow$ {T2, T3, T4}

E4 $\rightarrow$ {T2, T3, T4}

E5 $\rightarrow$ {T2, T3, T4}

others remain

E3 $\rightarrow$ {T1, T2, T3, T4}

E6 $\rightarrow$ {T1, T2, T3, T4}

The remaining two MAX branches do not need to be evaluated.
Therefore, the sin value

5, -1, 2

6, 4, 7 are pruned.

Total pruning

left branch: 2 leaves pruned

Right branch: 6 leaves pruned.

8 leaf nodes pruned.

∴ The Minimax Value is: 4

5. Delivery Route Optimization.

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

Hill climbing Strategy:

1. Start with current route.
2. Evaluate its total distance.
3. Generate neighboring routes.
4. select the neighbor with the lowest cost.
5. Move to that neighbor with lowest cost, improves solution.
6. Repeat until no neighboring route has a lower cost.

Route	cost
S1: $D \rightarrow A \rightarrow C \rightarrow B \rightarrow E \rightarrow D$	26 km
S2: $D \rightarrow B \rightarrow A \rightarrow C \rightarrow E \rightarrow D$	30 km
S3: $D \rightarrow A \rightarrow B \rightarrow E \rightarrow C \rightarrow D$	24 km
S4: $D \rightarrow E \rightarrow C \rightarrow B \rightarrow A \rightarrow D$	27 km
S5: $D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D$	23 km
S6: $D \rightarrow C \rightarrow B \rightarrow A \rightarrow E \rightarrow D$	29 km

According to the instruction, from S5, the best neighboring solution has a cost of 22 km, and all neighbors of that new solution have costs greater than 22 km.

step-by-step Hill climbing:

- Initial solution:

$$S5 = 23 \text{ km}$$

- A better neighboring solution is found:

$$23 \rightarrow 22 \text{ km}$$

∴

$$22 < 23$$

- The algorithm moves to the 22-km solution

*At the new solution:

All neighboring solutions > 22

Therefore, there is no further improvement.

$$\text{Final cost} = 22 \text{ km}$$

The algorithm stops because every neighboring solution has a higher cost.

$$\boxed{\text{MIN}(8, 5, 4) = 3}$$

Middle MIN

1. $(7, 5, 8)$

$\text{MAX} = 8$

2. $(6, 2, 3)$

$\text{MAX} = 6$

3. $(4, 0, -2)$

$\text{MAX} = 4$

$$\boxed{\text{MIN}(8, 6, 4) = 4}$$

Root MAX:

$$\boxed{\text{MAX}(4, 4, 3) = 4}$$

Alpha-Beta Pruning:

Using a left-to-right evaluation

left branch

For the second MAX node:

$(9, 1, 2)$

once 9 is found, the MIN Parent already has a value of 6 from the first branch

$\therefore 9 \geq 6$

the remaining values 1 and 2 do not need to be evaluated.

So:

1, 2 are Pruned

Right branch:

$\alpha = 4$

$\text{MAX}(1, 2, 3) = 3$

$\beta = 3$

$\therefore$

$\beta \leq \alpha$

$3 \leq 4$